Surprise!

We're not talking about ICT4D

- **Introduction**
  - Introduction
  - Overview of the Unit
  - Expectations
  - KWL
- **Lecture Part 1: Development, Social Transformation, & Complex Systems**
- **Exercise**
- **Lecture Part 1 Cont.**
- **Lecture Part 2: Why Design?**
- **Reflections**

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Introduction

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I'm Josh

- **Director, ICT4D Innovation and Scale, CRS**
- **Founder and Principal, Very Green Grass Co.**
- **PhD Researcher, Carnegie Mellon University**

- Formerly @CARE, founder and director of Research + Design team
- Formerly @UNICEF Lead at Innovations Lab Kosovo
- Formerly @UNICEF USA Strategic Partnerships and UNICEF Ventures
- Formerly @Teach for America Classroom Teacher, Newark Public Schools
- M.A. @Columbia School of Ed International Educational Development
- B.A. @Dickinson College Intellectual History

Overview

Module 1. Designing for the Social Good

1. **Part 1. Surprise! We're not talking about ICT4D.**
   Part 2. Complexity
2. **What's Design? Whose Design? Which Design**
3. **Sensing and Sensing Methods**
4. **Structuring the Unknown: Sensemaking**
5. **Creating Frame**

Module 2. Design Lab

1. **Part 1. Ideation: Enter the Sticky Note**
   Part 2. Concepting
2. **Designerly Ways of Doing**
3. **No Silver Bullets: Problematizing Design**

Unit Assignment

During the Design Lab, we'll hear from leading thinkers in the Digital Health field around Malaria campaign digitalization. You, in collaboration with your team, will be tasked with developing a strategy to researching and mapping the malaria campaign digitalization "system" in a given country, it's actors/actants and relationships, and propose an approach for making sense, framing, and facilitating the creation and articulation of alternative futures.

Expectations

1. **Read**
2. **Discuss**
3. **Be curious**

KWL

1. **5 minutes: What do you know about t4d/digital development and social design?**
2. **5 more minutes: What do you *want* to learn about ict4d/digital development and social design?**
3. **Email me at harvey.joshua@outlook.com**

Part 1.

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Development

What's *development*?

Development

Weber

- **Rationalization, and a shift toward calculability, predictability, efficiency, and control (Ritzer).**
- **States should build bureaucratic capacity and institutional rationality to manage social transformation.**
- **Bureaucracy as an ideal administrative form**
- **Rule-based governance**

Development

Parsons

- **Structural functionalism, Society as a system composed of interdependent subsystems performing necessary functions.**
- **Modernization = differentiation of roles and institutions**
- **Traditional societies are less differentiated and therefore less adaptive**
- ***Progress* requires institutional specialization (markets, bureaucracies, legal systems).**

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Development & Technology

Rostow

- **All societies move through five universal stages (tradition society, preconditions for “take off”, take off, drive to maturity, age of mass consumption).**
- **Progress is capital accumulation, industrialization, infrastructure investment, etc.**

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Development & Technology

Scott

- **High Modernism**

High Modernism

Scientific Ways of Knowing

Anthropocentric

Hegemonic

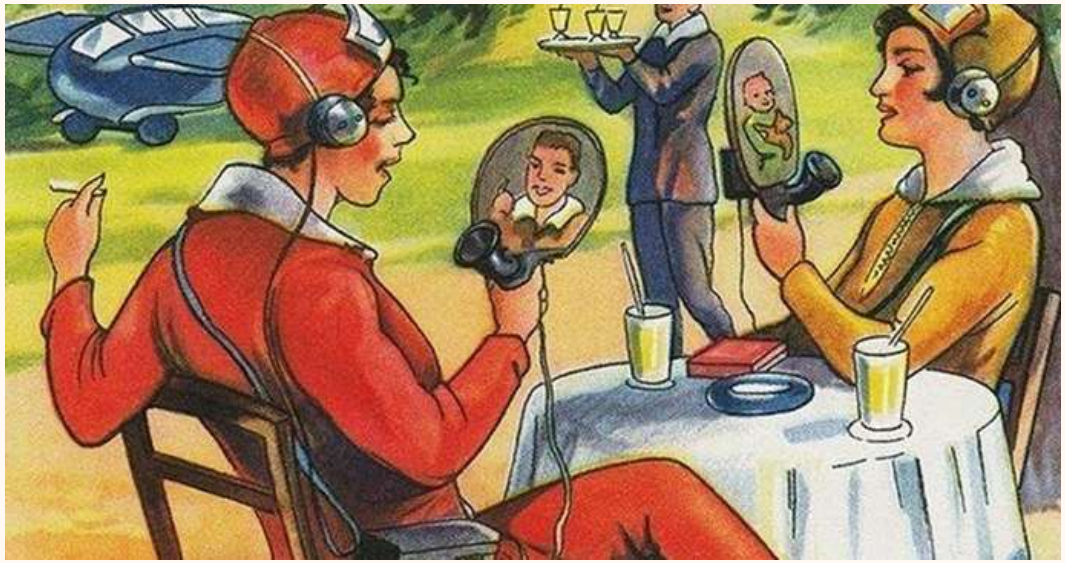
Capitalist Consumption

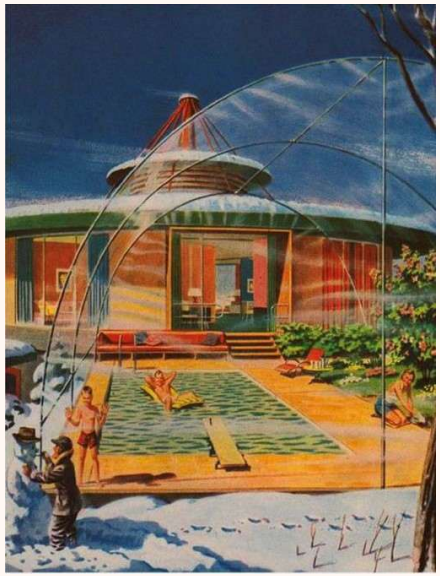
Atomization

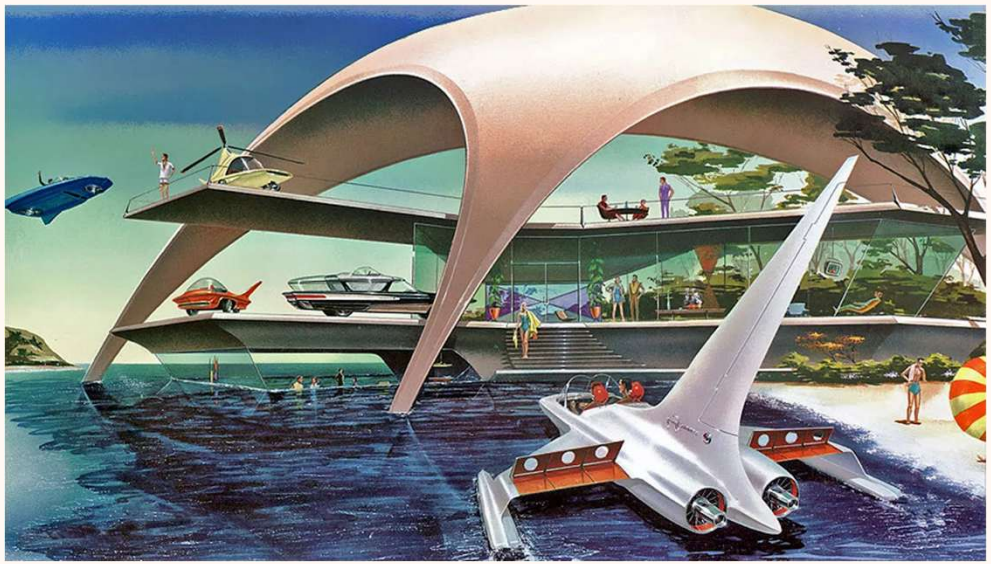
Individualism

Ahistorical

“Apolitical”







How's that going?

How it started...

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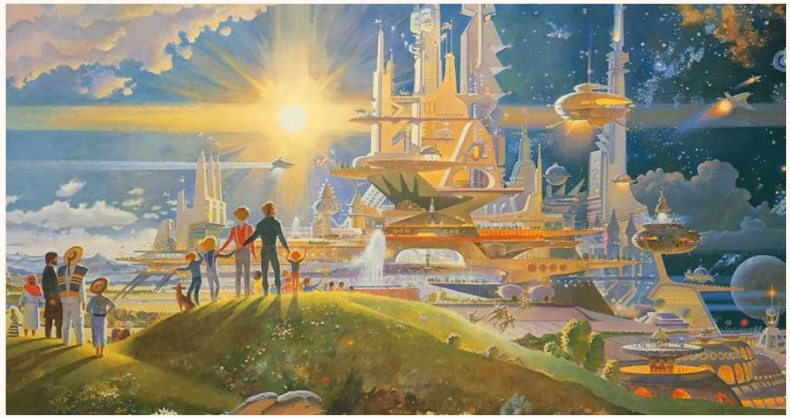

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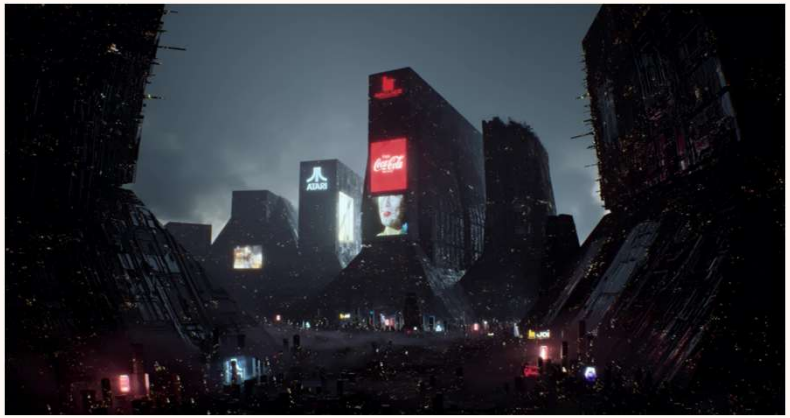

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Technē

Formal, codified, systematic knowledge oriented toward technical control and replicable production.

Development & Technology

What if development has *always* been about the application of technē – in its many forms – to the project of social transformation?

What if that's *a problem*?

Part 2.

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Post-Development

The emerging development – or "alternatives to development" – is one that acknowledges and seeks to reckon with radically large complex systems.

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Scientific Ways of Knowing

Anthropocentric

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High Modernism	**Post-Development**
Scientific Ways of Knowing	Alternate Worldviews
Anthropocentric	Holistic*
Hegemonic	Pluriversal
Capitalist Consumption	Degrowth
Atomization	Relationality
Individualism	Collectivism
Ahistorical	Contextualized
“Apolitical”	Situated and political

Technē

Formal, codified, systematic knowledge oriented toward technical control and replicable production.

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Mētis

Contextual, experiential, and embodied practical intelligence developed through lived engagement with particular situations and environments.

Technē

Operates and scales through *legibility*.

Mētis

Capacity to navigate complex systems.

Complex Systems

What's a system?

“Whenever I’m trying to help people understand what this word ‘system’ means, I usually start by asking: ‘Are you a part of a family?’ Everybody is a part of a family. ‘Have you ever seen in a family, people producing consequences in the family, how people act, how people feel, that aren’t what anybody intends?’ Yes. ‘How does that happen?’ Well… then people tell their stories and think about it. But that then grounds people in not the jargon of ‘system’ or ‘systems thinking’ but the reality – that we live in webs of interdependence.”

Peter Senge

Complex Systems

What's a *complex* system?

Exercise

Break into groups. Each group will be assigned one concept. Read the relevant section and prepare a 60 second summary for the group.

Group 1. Interconnected and Interdependent Elements

Group 2. Feedback Processes

Group 3. Emergence

Group 4. Nonlinearity

Group 5. Adaptive Agents

Sociotechnical Systems

What's a *sociotechnical* system?

> “The entanglement and emergent phenomena associated with the interactions/negotiations between and interconnected changes within self-reinforcing domains of technology, the economy, institutions, behaviour, and cultural systems”

- Savaget, Geissdoerfer, Kharrazi, Evans

Sociotechnical Systems

Complex systems that involve interactions between humans, technology, and their environment.

Note: The term (and the related Socio-technical Systems Theory) holds different connotations in different traditions. In some – namely organizational development/industrial organizational psychology, in this context, t*echnology* is used more in the Greek "techne" – the application of specialized knowledge.

A Latour Detour

Elaborated by, among others, Latour.

> Society and technology are not two ontologically distinct entities but more like phases of the same essential action.”

Latour

> **Artifacts can be deliberately designed to replace human action and constrain and shape the actions of other humans.”**

Latour

> Technologies that are so commonplace that we don't even think about them can shape the decisions we make, the effects our actions have, and the way we move through the world."

Latour

Sociotechnical Systems

In summary, we delegate our duties and values to technologies (here largely meaning material artefacts), technologies in turn act in sociotechnical systems to influence human behaviours.

ICT4D

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Intervening in Complexity

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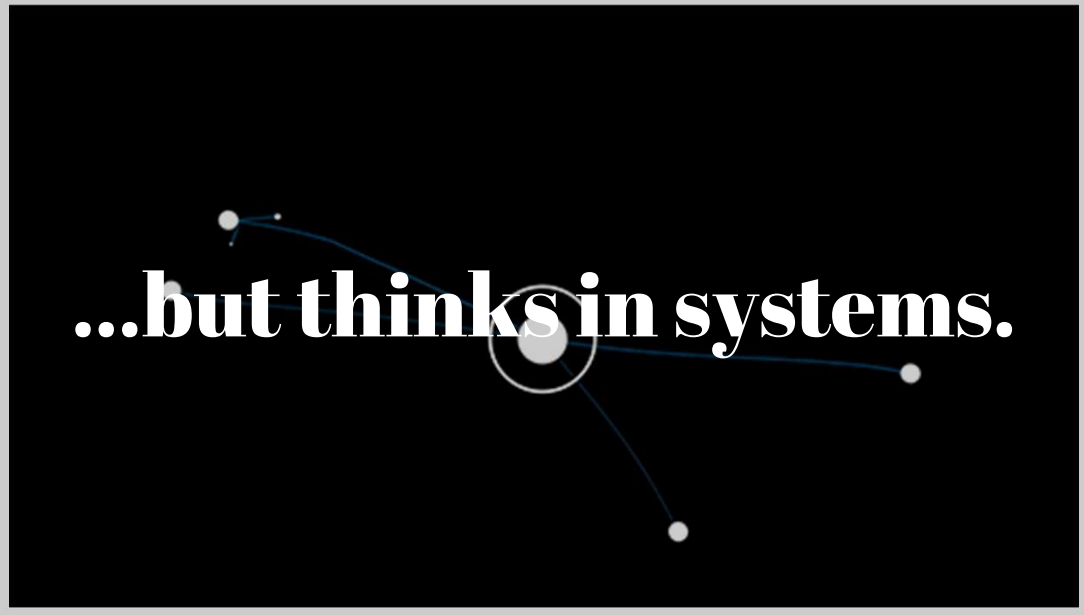
Design learns by trying.

Design centers people...


...but thinks in systems.

Design shifts frames.

Assignment 1

For next class, write a one page (500 words max) sharing your observations (personal, professional, academic) on how high modernist values/a preoccupation with *technē* manifests in the international development and humanitarian space. Use concrete examples if possible.

Thanks

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